



HENRY CONTROL WORKS

MACHINE LOGIC GROUP · CONVERSION SYSTEMS DIVISION

HCW-7 DETERMINISTIC TAPE PROCESSOR

Expanded Operator & Programming Reference

Core operation, application patterns, tape protocols, qualification procedure, and service diagnostics.

CONTROLLED COPY

Training exercises are issued separately. This reference remains at the operator station.

EMPLOYEE ISSUE RECORD

ASSIGNED TO

WORKSTATION / SHIFT

DOCUMENT

HCW-7/OPR-02

REVISION

2026.08A

OPERATOR ASSIGNMENT

Your station converts specifications into machines

A customer supplies a transformation: one binary input word must become one binary output word. You define the HCW-7’s rules, inspect one run, then qualify the source against the full supplied batch.

01

Read

Identify the required input-to-output behavior.

02

Write

→ Enter states and conditional rules in the source editor.

03

Load

→ Place one chosen input on the tape.

04

Inspect

→ Run the simulator and watch the machine cycle.

05

Qualify

→ Submit against every visible test case.

THE TWO GREEN LIGHTS DO NOT MEAN THE SAME THING

ACTION	WHAT IT PROVES	WHAT IT DOES NOT PROVE
RUN SIMULATION	The current source produced a result for the input you loaded.	That the source handles any other input.
SUBMIT SOLUTION	The source halted with the expected output for every supplied test case.	That the machine is efficient, minimal, or correct beyond the tested domain.

OPERATOR STANDARD

- Inspect a failure before editing at random.
- Read the active state and current symbol together.
- Use qualification after the local run makes sense.
- Optimize only after the behavior is correct.

FOREMAN’S NOTE

If the machine is correct only on the input you watched, it is not correct. It is acquainted with one example.

A

PROCESSOR

Tape, head, state register, rule selector, and halt circuit.

B

SOURCE

State blocks and one conditional transition per read symbol.

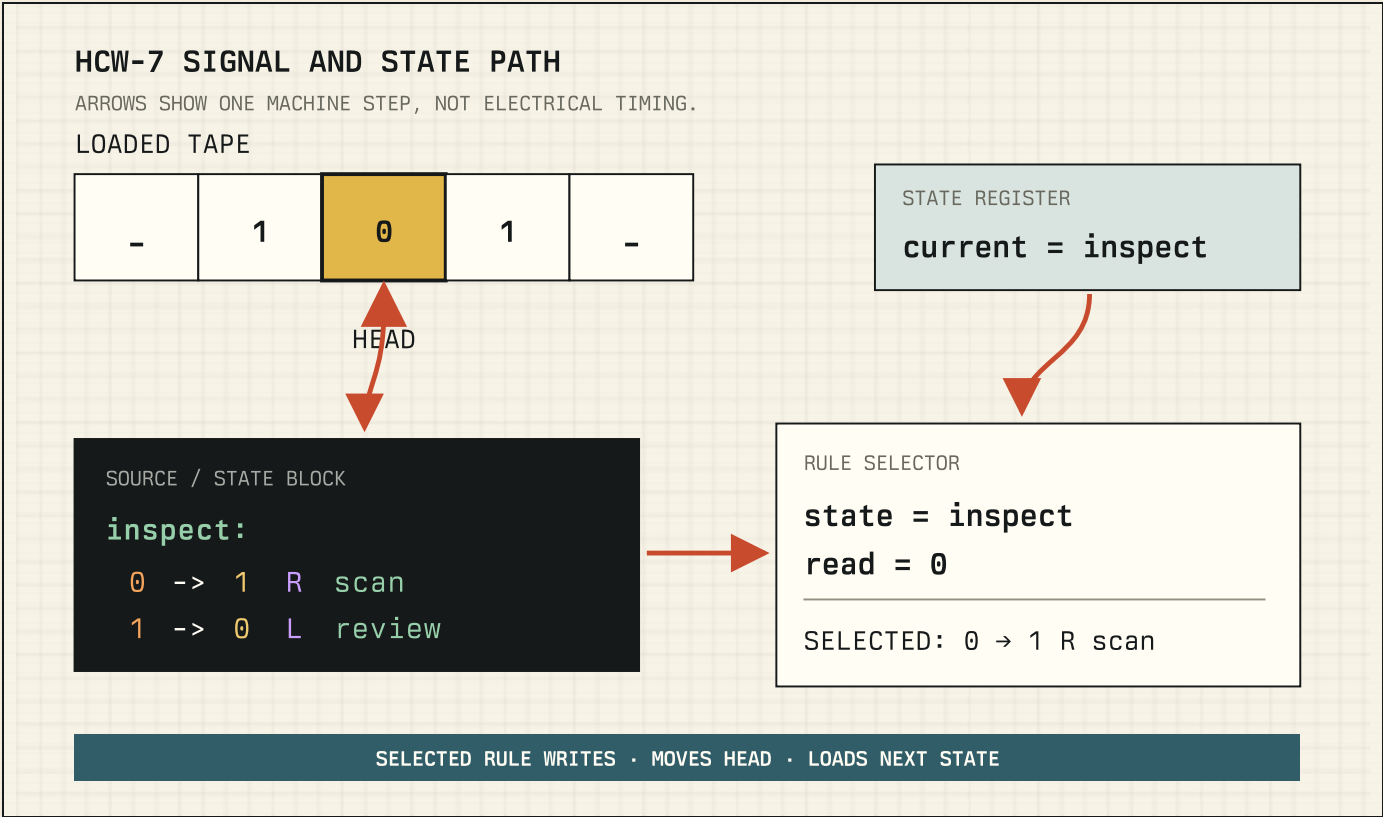
C

QUALIFICATION

Expected outputs, verdicts, and post-pass efficiency measures.

HARDWARE MAP

One current cell, one current state, one selected rule



TAPE

An unbounded row of cells. Each cell contains 0, 1, or blank -. Input begins at cell 0.

HEAD

Points to one current cell. The machine can read and write only that cell during the current step.

STATE

Selects one named group of rules. Every run begins in q0.

OUTPUT IS NOT A PORT. After HALT, the system derives output from the final nonblank span of tape. See Sheet C-1.

EXECUTION SEQUENCE

Each step is one complete five-part transaction

Rules in a state are alternatives. The processor does not run them from top to bottom. It chooses the rule whose read symbol matches the current cell.

01

Read

Sample the symbol under the head.

02

Select

→ Find the matching rule in the current state.

03

Write

→ Replace the current cell's symbol.

04

Move

→ Shift the head left, right, or stay.

05

Next

→ Load the destination state or halt.

ONE TRANSACTION UNDER INSPECTION

-	1	0	1	-	-	-
---	---	---	---	---	---	---

HEAD

0	→	1	R	scan
---	---	---	---	------

-	1	1	1	-	-	-
---	---	---	---	---	---	---

HEAD

Result of the step: the current cell changed from 0 to 1, the head moved one cell right, and scan became the current state.

SELECTION PRIORITY

An exact rule for 0, 1, or _ takes priority over the fallback rule *.

```
scan:  
* -> ~ R scan  
_ -> _ S HALT
```

On blank, the exact _ rule is selected. On 0 or 1, the fallback rule is selected.

NO MATCHING RULE

The processor becomes STUCK. It does not skip to another state.

WRITE PRECEDES MOVE

The symbol is written in the cell the head occupied at the start of the step.

MOVE PRECEDES NEXT READ

The destination state reads whichever cell becomes current after movement.

HALT ENDS THE CYCLE

A transition to HALT still performs its write and movement before stopping.

MACHINE CYCLE

03

SHEET A-2

PROGRAMMING REFERENCE

Source is organized into named state blocks

```
# comments begin with #
q0:
  0 -> 1 R inspect
  1 -> 1 R inspect

inspect:
  0 -> ~ L review
  1 -> 0 S HALT
  _ -> _ S HALT
```

■ state name ■ read symbol ■ write symbol ■ movement

REQUIRED START STATE

Every source must declare **q0**. Every simulation begins there.

STATE NAMES

Begin with a letter, then use letters, digits, or underscores. Names are case-sensitive except **HALT**, which is recognized in any letter case and formatted as **HALT**.

ONE CONDITION EACH

A state may define at most one rule for each read symbol. Duplicate conditions are a compile error.

CANONICAL RULE FORM

READ	→	WRITE	MOVE	NEXT STATE
READ 0 1 _ * Condition tested against the current cell.		WRITE 0 1 _ ~ Symbol stored in the current cell.	MOVE L S R Direction applied after the write.	NEXT name HALT State loaded after movement.

WHITESPACE AND COMMENTS

Indentation is for readability. Blank lines are ignored. Anything after # on a line is a comment.

EDITOR COMPLETION

The source editor can complete separators, preserve symbols, movements, state names, and **HALT**. Completion assists entry; the grammar remains authoritative.

DO NOT READ RULES AS A SCRIPT. Inside a state, rows are conditions. Only one row is selected per step.

SOURCE

OPERAND REFERENCE

Read conditions and write actions use different alphabets

READ SYMBOLS			WRITE SYMBOLS		
TOKEN	MATCHES	USE	TOKEN	ACTION	USE
0	Current cell contains 0	Exact condition	0	Store 0 in current cell	Literal write
1	Current cell contains 1	Exact condition	1	Store 1 in current cell	Literal write
-	Current cell is blank	Exact condition	-	Erase current cell to blank	Literal write
*	Any symbol without a matching exact rule	Fallback condition	~	Write back the symbol just read	Preserve current cell

* IS NOT “READ EVERYTHING IN SEQUENCE.” It is one fallback condition. ~ is not a tape symbol; it means preserve the symbol selected by the current rule.

L MOVE LEFT Head index decreases by one.	S STAY Head remains on the written cell.	R MOVE RIGHT Head index increases by one.
---	---	--

EXACT RULE OVERRIDES FALLBACK			HALT DESTINATION
CURRENT SYMBOL	RULES PRESENT	SELECTED CONDITION	The current rule still writes and moves. The machine then stops instead of selecting another state.
0	0 and *	0	
1	0 and *	*	
-	- and *	-	

CONSOLE OPERATION

The simulator is an inspection bench, not a grader

1. INPUT

Enter the binary word you want to inspect. Use **LOAD** to place it on the tape and reset the machine.

2. SOURCE

Edit state blocks. A compile error identifies the first source line the processor cannot accept.

3. TRANSPORT

Step backward, play or pause, step forward, and reset the recorded run. Speed changes playback, not machine behavior.

PLAYBACK RECORD

STEP	STATE	READ	WRITE
14	scan	1	0

The state diagram highlights the active state and selected rule. The tape marks the current cell. Read those surfaces together.

TRANSPORT CONTROLS

◀	Previous recorded step
▶ /	Play or pause
▶	Next recorded step
RESET	Return to loaded input
1x...1024x	Simulation playback speed

OUTPUT READOUT

The ordinary output panel abbreviates long values when needed. **FULL OUTPUT** opens the complete final word for inspection and copying.

OUTPUT

101101...001

FULL OUTPUT

RUN SIMULATION

Executes the current source once on the loaded input and records the steps for inspection.

It does not run the test table.

THE GRAPH IS A DEBUGGING VIEW. It shows the machine you wrote. It does not tell you which machine the specification requires.

CONSOLE

FINAL TAPE INTERPRETATION

Output is the contiguous span between the outermost nonblank cells

After the processor reaches **HALT**, leading and trailing blank cells are discarded. Every cell between the leftmost and rightmost nonblank cell becomes the output word.

FINAL TAPE EXAMPLE

-	-	1	0	1	1	-	-	-
---	---	---	---	---	---	---	---	---

ignored blank region

OUTPUT = 1011

ignored blank region

POSITION

The output may appear anywhere on the tape. Absolute cell indices are not judged.

HEAD

The final head position is not judged once the machine halts.

EMPTY OUTPUT

If every cell is blank at halt, the output is the empty word.

QUALIFICATION TABLE

INPUT	EXPECTED OUTPUT	MACHINE OUTPUT	RESULT
0	1	1	PASS
1	10	10	PASS
101	110	100	FAIL

Illustrative values only. Each level supplies its own transformation and test cases.

SUBMIT SOLUTION

Compiles the current source, runs every supplied test case, and compares each final output to its expected output.

Any compile error, stuck run, timeout, wrong output, or incomplete batch prevents acceptance.

SPECIFICATIONS DESCRIBE WORDS, NOT CELL ADDRESSES. Do not preserve a position the grader does not observe unless your own machine needs it.

DIAGNOSTICS

Read the first broken contract

STATUS	MEANING	OPERATOR ACTION
COMPILE ERROR	The source does not match the accepted grammar or declares an invalid machine.	Read the reported line. Check the state declaration, arrow, symbols, movement, and destination slots in that order.
STUCK	The current state has no exact rule and no fallback rule for the current symbol.	Inspect the recorded state and read symbol. Add the missing condition only if the specification requires that path.
TIMEOUT	The machine exceeded the permitted step count without halting.	Find the repeating state, head region, or tape pattern. Repair the loop or prove that it makes progress toward a boundary.
FAIL	The machine halted, but its final nonblank span differs from the expected output.	Compare the first differing output position, then trace backward to the write that created or erased it.
PASS	The machine halted with the expected output for that test case.	Continue through the batch. One passing row is evidence about one input.

TRACE-FIRST FAULT ISOLATION

1. Load the failing input from the qualification table.

2. Run the simulation and move to the last correct step.

3. Record the active state, symbol read, selected rule, and next head position.

4. Decide whether the wrong decision came from the state, condition, write, movement, or destination.

5. Edit one cause, then replay the same input before resubmitting.

DO NOT CONCEAL FAULTS

A wildcard can hide missing exact cases.

Extra cleanup can destroy correct output.

A halt can freeze a locally correct but incomplete tape.

A faster loop is still the wrong loop.

STATE

Which group of conditions was active?

READ

Which symbol selected the rule?

HEAD

Which cell becomes current next?

RANDOM EDITING DESTROYS EVIDENCE. Preserve the failing input and the last correct step until you can name the broken decision.

POST-QUALIFICATION ENGINEERING

Correct machines are compared on states and cumulative steps

The public frontier contains solutions that are not beaten on both ranked measures. A machine can trade a smaller state count for more work, or more states for fewer total steps.

STATES

The number of distinct named states in the accepted source.

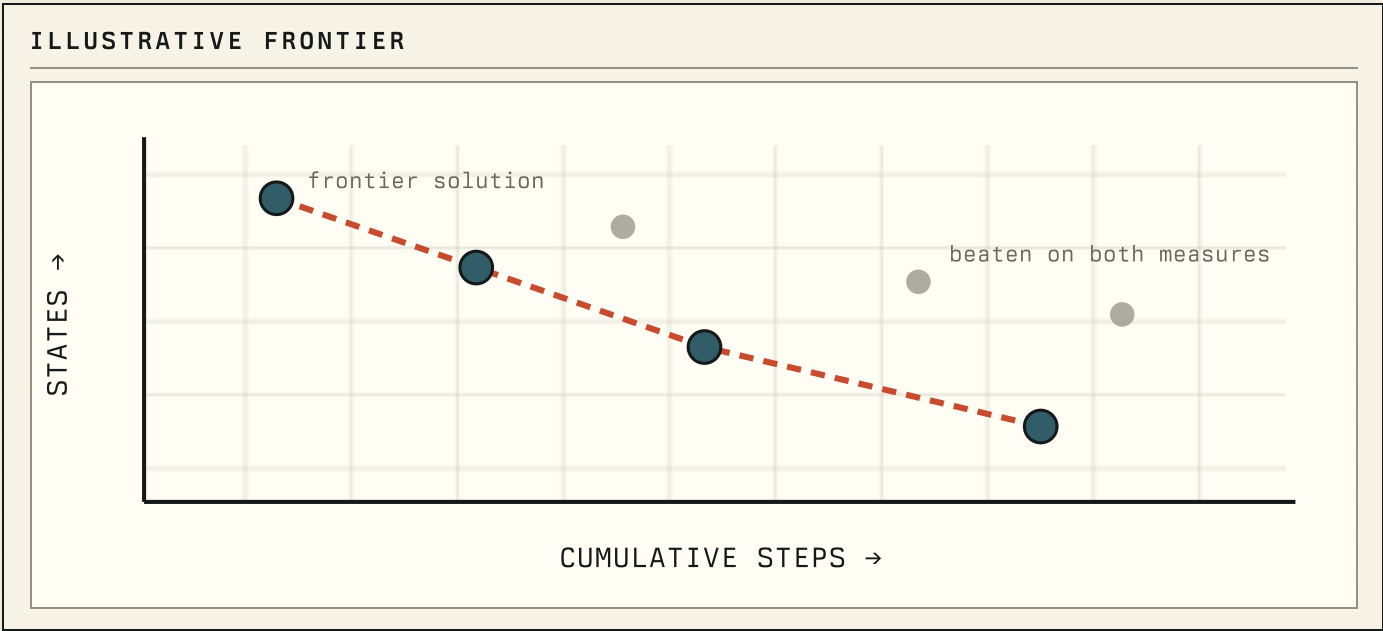
Fewer states usually means a smaller control plan. It does not guarantee fewer steps.

STEPS

The sum of executed machine steps across the qualification test batch.

This is cumulative work, not the longest single run.

METRICS



FIRST

Pass every supplied test. An invalid machine has no useful efficiency score.

THEN

Choose one measure to reduce and watch what happens to the other.

KEEP

Save distinct tradeoffs. The smallest machine and fastest machine may differ.

RULE COUNT IS DIAGNOSTIC, NOT RANKED. The public comparison uses states and cumulative steps.

ADVANCED CONTROL PLANNING

APPLICATION NOTE ONE

A state can remember one finite fact

The active state is more than a phase label. It can preserve a small fact after the head leaves the cell where that fact was read.

BRANCH INTO MEMORY

ENTRY FRAGMENT TRANSIT AND BOUNDARY RULES OMITTED

```
inspect:
  0 -> 0 R carry0
  1 -> 1 R carry1
```

After this branch, the state name preserves the symbol. The source cell no longer needs to remain under the head.

MEMORY MEDIUM

USE STATE	USE TAPE
Few possible values	Many possible values
Fact survives one phase	Fact must survive revisits
No spatial lookup needed	Location is part of the method

PRODUCT-STATE CONSTRUCTION

When a phase and a remembered bit matter at the same time, give each combination its own state.

PHASE	REMEMBER 0	REMEMBER 1
OUTBOUND	out0	out1
RETURN	return0	return1

This trades more states for simpler tape logic. Merge combinations only when they truly select identical rules.

STATE MEMORY IS VOLATILE. A transition to a state with no encoded fact discards that fact immediately.

ONE BIT

Two states encode two possible remembered values.

TWO FACTS

The combinations multiply. Keep only combinations the machine can reach.

REVIEW

Ask whether the remembered fact changes the next selected rule.

ADVANCED CONTROL PLANNING

APPLICATION NOTE TWO

Carry a symbol by separating capture, transit, and discharge

Read the source symbol once, branch into the matching memory state, cross the intervening tape, then let the arrival rule write the remembered value.

CAPTURE Read the source cell.	BRANCH → Enter <code>carry0</code> or <code>carry1</code> .	TRANSIT → Move without losing the state.	ARRIVE → Detect the agreed boundary.	DISCHARGE → Write the value encoded by the state.
---	---	--	--	---

CAPTURE FRAGMENT

VALID SOURCE FRAGMENTNO DESTINATION SHOWN

```
remember:
  0 -> 0 R carry0
  1 -> 1 R carry1
```

DISCHARGE FRAGMENT

BOUNDARY-ONLY RULESTRANSIT RULES OMITTED

```
carry0:
  _ -> 0 L return

carry1:
  _ -> 1 L return
```

CONTROL PROOF	
QUESTION	REQUIRED ANSWER
What fact is carried?	The state name identifies exactly one source value.
What preserves it?	Every transit rule stays inside the corresponding carry state.
What ends transit?	An exact boundary rule overrides the interior scan behavior.
What happens after writing?	The next state establishes a new phase and drops the carried fact.

DO NOT RE-CAPTURE BY ACCIDENT. If a transit state branches on unrelated interior symbols, the carried value can change before arrival.

The fragments above describe one control pattern. They do not specify where a destination belongs or which transformation a level requires.

TAPE ORGANIZATION

APPLICATION NOTE THREE

Use blank separators only with a stated region invariant

A blank cell may sit outside the data, divide two work regions, or be the remains of erased data. The symbol alone does not tell the machine which meaning applies.

-	1	0	1	-	1	1	-	0	1	-
INPUT REGION				WORK REGION			AUX REGION			

REGION CONTRACT

- Each active region contains only the symbols its phase expects.
- The separator stays blank until its last consumer has crossed it.
- The state identifies which boundary the head is seeking.
- Every write preserves the contract or deliberately enters a new one.

WHY POSITION MATTERS

The same `_` can mean “turn around” during an outbound sweep and “leave the active area” during cleanup. Direction and phase provide the missing context.

A coordinate is not an intrinsic origin. The loaded input merely begins at the interface’s input-start coordinate.

AMBIGUITY AUDIT		
SITUATION	RISK	REQUIRED CONVENTION
Region may be empty	Its two boundaries collapse into a run of blanks.	Use another representation or a phase that already knows its exact neighborhood.
Interior cell is erased	A sweep may mistake the hole for the region end.	Avoid holes, or use a control plan that expects them.
Several regions exist	The next blank may be the wrong delimiter.	Name the destination in the current state and preserve the crossing count.

A DELIMITER IS A PROTOCOL, NOT A COLOR. It works only because every reachable state agrees on what lies to either side.

TAPE ORGANIZATION

APPLICATION NOTE FOUR

Give each leg of a shuttle its own control phase

A shuttle repeatedly moves between regions. The same interior symbols appear on both legs, but the head direction and boundary response differ.

OUTBOUND Scan toward the destination region.	TURN Detect its boundary and establish the return phase.	INBOUND Scan back toward the source region.	RE-ENTER Land on the intended source-side cell.
--	--	---	---

RIGHTWARD INTERIOR

SCAN BODY

BOUNDARY RULE OMITTED

```
to_work:
0 -> ~ R to_work
1 -> ~ R to_work
```

LEFTWARD INTERIOR

SCAN BODY

BOUNDARY RULE OMITTED

```
to_source:
0 -> ~ L to_source
1 -> ~ L to_source
```

LANDING CALCULATION

A boundary rule reads the current blank, writes, moves, then changes state. The next state reads the cell after that movement. Design the turn from that exact landing cell.

STATE NAMING

to_work and to_source expose direction and purpose. Names such as q7 are legal but make trace diagnosis harder.

ONE SCAN STATE SHOULD NOT GUESS ITS DIRECTION. Reuse it only when the symbol read fully determines whether the head must move left or right.

SHUTTLE COST

If the distance between regions grows while the shuttle repeats, cumulative steps can rise quickly. After correctness, consider whether one full sweep can carry more useful work.

CONTROL PROOF

APPLICATION NOTE FIVE

Every repeated sweep needs visible progress

A phase may revisit the same region many times. Termination comes from a quantity that changes irreversibly on each pass, not from the hope that the next pass behaves differently.

ENTER

Start from a known side.

SCAN

→ Preserve the current region invariant.

CHANGE

→ Consume or resolve one unit.

TURN

→ Detect a boundary and reverse phase.

TEST

→ Halt or begin a strictly smaller pass.

PROGRESS CERTIFICATES

- The unresolved suffix loses one symbol.
- A frontier moves one cell and never moves back.
- One marker is consumed and never recreated.
- A finite work region becomes permanently settled.

CYCLE CERTIFICATE

The processor is deterministic. If the same state, tape contents, and head position recur together, every later step repeats forever.

A familiar-looking local window is not enough; the complete reachable configuration matters.

TURN AUDIT

AT EACH BOUNDARY	VERIFY
Read	The boundary symbol is the one the phase expects.
Write	The separator or frontier remains valid for the next phase.
Move	The head lands inside the intended next region, not on the far side.
Next state	The new rules match the cell reached after movement.

NAME THE DECREASING QUANTITY. If you cannot point to what becomes smaller or more settled after a pass, termination is not yet designed.

SHORTEST INPUT

Does the first pass still make progress?

UNIFORM INPUT

Can repeated symbols hide a missing turn?

BOUNDARY CASE

Does a pass stop one cell too soon or too late?

APPLICATION 5

TAPE ORGANIZATION

APPLICATION NOTE SIX

Choose whether work may destroy the original input

In-place work often uses fewer cells. A preserved copy makes old facts easier to recover. The output contract decides what must remain at halt, not which method is inherently correct.

PLAN	ADVANTAGE	COST	MAIN RISK
Destructive	Reuses the loaded input region.	Overwritten facts may need state memory or reconstruction.	Changing a symbol before its last consumer reads it.
Nondestructive	Keeps a stable reference copy.	Copying and shuttling add states or steps.	Forgetting to remove obsolete regions before halt.
Hybrid	Preserves only the facts needed later.	Requires a precise lifetime plan.	Keeping too little or cleaning too early.

PREPARE

Establish separators and any stable copy.

WORK

Transform one region under a stated invariant.

CONSOLIDATE

Place every required final bit in one span.

CLEAN

Erase obsolete nonblank cells, then halt.

CLEANUP BODY

INTERIOR FRAGMENT

STOP RULE OMITTED

```
cleanup:
  0 -> _ R cleanup
  1 -> _ R cleanup
```

OBSERVED OUTPUT

At halt, the validator reads the entire span from the leftmost to rightmost nonblank cell. A surviving scratch island expands that span, and a blank gap inside it remains part of the observed tape span.

SCRATCH SPACE HAS A LIFETIME. Decide when each temporary cell is created, which phase last reads it, and when it becomes safe to erase.

TAPE ORGANIZATION

APPLICATION NOTE SEVEN

Write a spatial protocol before writing transition rules

A protocol states what each region means, where the head may be during each phase, and which facts remain true after every transition.

EXAMPLE WORKING FORM

-	1	0	1	-	1	1	-	0	1	-
INPUT REGION				WORK REGION				AUX REGION		

This is notation for planning, not a required level format or a complete machine.

PROTOCOL FIELDS

REGION

Which symbol sequences are legal here?

HEAD

Where can the current phase begin and end?

FRONTIER

Which boundary moves, and in which direction?

LIFETIME

When may a delimiter or temporary symbol be erased?

STATE INVARIANTS

- In **to_work**, the head moves right inside an established region.
- In **return0**, the head moves left while the state remembers 0.
- In **cleanup**, every visited data cell becomes blank.
- In a turn state, the head is adjacent to the boundary it just crossed.

BOUNDARIES ARE RELATIONAL. The machine recognizes a useful blank because its path and state place the head beside the region that blank bounds.

PROTOCOL REVISION RULE

If one transition breaks the stated region form, either the transition is wrong or the protocol is incomplete. Revise the diagram before distributing patches across unrelated states.

ENGINEERING REVIEW

APPLICATION NOTE EIGHT

Review the control plan before optimizing the source

A useful review names the machine’s memory, boundaries, progress measure, and final cleanup. It does not begin by rearranging rules for a smaller score.

CONTROL REVIEW

01

Which facts live in state, and which live on tape?

02

What does each work region contain in every phase?

03

How does the head distinguish the boundary it seeks?

04

What irreversible change proves each repeated sweep progresses?

05

Which temporary cells must disappear before halt?

PROBE SET

Empty input	When the level domain permits it
One symbol	Both 0 and 1
Uniform word	All zeroes and all ones
Alternating word	Repeated boundary changes
Leading zeroes	When the input is a word, not canonical integer
Shortest regions	Separator-delimited structures

OPTIMIZATION ORDER

1 / SPECIFY

State the observed transformation and domain.

2 / PROVE

Name invariants and progress.

3 / QUALIFY

Pass every supplied test case.

4 / MEASURE

Inspect states and cumulative steps.

5 / TRADE

Change one ranked measure deliberately.

STATE

Can two control combinations select identical rules?

STEP

Can one pass replace repeated shuttles?

RISK

Did the change weaken a boundary or cleanup invariant?

CORRECTNESS FIRST.

A smaller source that fails one supplied case is not an efficient machine.

STATION COPY

HCW-7 quick reference

SOURCE SKELETON

```
q0:
  READ -> WRITE MOVE NEXT

next:
  READ -> WRITE MOVE HALT
```

MACHINE STEP

READ → SELECT → WRITE → MOVE → NEXT

Every run starts in q0. A transition to HALT writes and moves before stopping.

TOKENS

Read	0 1 _ *
Write	0 1 _ ~
Move	L S R
Next	state HALT

PRIORITY

Exact rule before fallback.

* matches only when no exact 0, 1, or _ condition exists in the current state.

OUTPUT CONTRACT

At halt, output is the tape span from the leftmost to rightmost nonblank cell. Outer blanks, absolute position, and final head position are ignored. An all-blank tape produces the empty word.

CONSOLE ACTIONS

RUN SIMULATION inspects one loaded input. SUBMIT SOLUTION qualifies the source against every supplied test case.

FAILURE QUESTIONS

1. Which state was active?

2. Which symbol was read?

3. Which rule was selected?

4. What cell becomes current next?

RANKED MEASURES

States counts named control states. Steps sums executed steps across the qualification batch.

OPERATOR RETENTION LINE

The machine does exactly what the selected rule says.

Your work is deciding which rule should be selectable in every reachable situation.

HCW

7

REFERENCE